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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026, 07:59:59 AM MYT

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1. Team & Project Overview

Team Name	Nexus
Project / Solution Name	NEXUS / AI Career Navigator
Name (Leader)	XIE YIFENG
Email & Phone (Leader)	ait2609068@xmu.edu.my | Phone: 1114401090
University Affiliation(s)	Xiamen University Malaysia (XMUM)
Live Prototype URL	https://ai-career-navigator-xyf20070924.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/xieyifeng24/AI-Career-Navigator (public)

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

The browser collects structured answers and computes the base report locally from validated occupational snapshots. The same report feeds the career constellation, Skill DNA, learning plan, What If and SDG views. Browser persistence supports refresh recovery. FastAPI serves allowlisted assets and knowledge/regional JSON. Data-refresh modules retrieve official sources or serve reviewed extracts; failed updates retain a validated snapshot. With explicit consent, a separate backend path sends bounded answers and approved context to RunPod's Qwen vLLM endpoint. Server validation checks schemas and allowed evidence/source/action IDs; failure leaves the local report available. Anonymous signed sessions, same-origin/CSRF checks, SQLite quotas and temporary owner-scoped caches bound optional inference. SQLite stores keyed call metadata, not a profile database. Replit hosts the Python app; GitHub hosts source and CI/data-check workflows. A configured workflow is not proof that scheduled refresh has run. Historical radar/matching endpoints coexist with the default combined path; the historical RunPod echo handler is not the deployed inference worker. See Appendix A.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
HTML / CSS / JavaScript ES modules	Native responsive UI; local evidence analysis; no frontend framework	Web standards; project code Apache-2.0
Python 3.12 / SQLite	FastAPI server; standard-library JSON/SQLite quota ledger	PSF / SQLite public domain
FastAPI 0.156.4 / Pydantic 2.14.1	HTTP API and bounded schema validation	MIT
HTTPX 0.28.1 / Uvicorn 0.54.0	External requests / ASGI serving	BSD-3-Clause
Beautiful Soup 4.15.0 / python-dotenv 1.2.4	Legacy public-page parsing / environment configuration	MIT / BSD-3-Clause
All remaining requirements.txt pins	Complete declared Python inventory in Appendix C	Component terms listed there

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod OpenAI-compatible vLLM API	Optional consent-based model explanations; server-only credential	credit tier unverified
ESCO official API / O*NET official files and pages	Knowledge snapshot imports; O*NET authenticated Web Services 2.0 NOT configured	Public access; source licences apply
DOSM storage/data catalogue / World Bank Indicators API	Bounded regional evidence refresh	Public/free access; no API key shown
GitHub raw content	Validated published knowledge/regional snapshot retrieval	Public; usage limits apply
arXiv cs.AI RSS (legacy radar endpoint)	Recent research context only; not an employment forecast	Public RSS; article-specific rights

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Public FastAPI deployment; cloudrun config; port 8000	Live URL verified; account region not verified
RunPod Serverless vLLM	Optional Qwen inference; documented worker v2.29.0	Provisioned use in repository region not verified
GitHub	Public source, pull requests, CI and data-check workflow code	Public repo; Actions execution/limits not independently verified
Browser localStorage / local SQLite / JSON	Client drafts; server keyed quotas; knowledge snapshots	Local application storage, not a managed cloud profile database

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen3-4B-Instruct-2507	Qwen; served through RunPod vLLM	Pretrained; no fine-tuning or training declared in inspected implementation	Apache-2.0 + RunPod terms
Revision cdbee75f17c01a7cc42f958dc650907174af0554	Repository provision record; worker v2.29.0	Historical successful requests recorded; current production inference not independently verified	Model Apache-2.0; image components have own terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
O*NET 31.0; 20 records	onetcenter.org/database.html	Public; selected tasks/skills	CC BY 4.0; attribution retained	N
ESCO; 7 records	esco.ec.europa.eu	Public classification; API snapshot	European Commission reuse terms; attribution	N
DOSM / MASCO; MyCOL; BLS; World Bank	Official URLs in regional/current.json and Appendix C	Public statistics/classification; mixed API and reviewed extracts	Publisher-specific terms; attribution retained	N
arXiv RSS / UN SDG definitions / learning-resource links	rss.arxiv.org; sdgs.un.org; Microsoft/Google/Python documentation	Public references; no full-paper/course redistribution	Source-specific rights	N
Fictional personas / self-authored questionnaire and SDG mappings	Project source and acceptance fixtures	Synthetic / team-authored; no claimed psychometric validation	Project code: Apache-2.0; original team assets	N (fixtures); live answers may be personal

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
OpenAI Codex (exact model/version not established)	Implementation/testing/docs in project records; current submission preparation	Major
ChatGPT (version not recorded)	Planning, analysis, drafting and coding support	Major (team-confirmed)
DeepSeek (version not recorded)	AI-assisted coding, analysis and debugging; not the runtime model	Major (team-confirmed)

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

The team used ChatGPT, OpenAI Codex and DeepSeek to assist with planning, coding, testing, documentation and debugging. Human team members reviewed the project source and generated outputs, used automated checks and fictional-persona flows, and kept occupational evidence separate from model interpretation. The final team remains responsible for originality, accuracy, licensing, security and the signed submission. Optional AI responses are schema-checked and cannot replace the local baseline.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?
☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Hosting/platform	Public prototype URL successfully opened; actual sponsor credits not confirmed
RunPod	Serverless GPU inference	Provisioned endpoint/model and genuine evidence recorded in repository; actual sponsored tier not confirmed
Workshops / mentoring / other sponsor data	Other workshop/mentoring benefits not included in this claim	Replit and RunPod usage confirmed. No further sponsor-provided services asserted.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides the public deployment; the FastAPI app could be hosted on another compatible Python platform. RunPod provides optional model interpretation and can be replaced by a compatible server-side inference provider; the deterministic core report works without inference. Platform usage is not proof of sponsor-credit usage. Genuine existing evidence is referenced in Appendix D; no new account usage or billing evidence is fabricated.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The questionnaires process personal context and unverified self-reports. Competition demonstrations should use fictional profiles. Core answers, optional story and reports remain in browser state/localStorage and can persist across refreshes; users should use reset and browser storage controls on shared devices. Core analysis needs no account or model request. Optional AI requires explicit consent and sends a bounded payload through FastAPI to RunPod; the current AI-practice guide sends twelve direction answers and server-owned evidence, excluding the optional story. Other legacy AI routes have different synthetic-profile contracts. No server profile database is implemented. Temporary owner-scoped result caches expire; SQLite stores keyed network/session identifiers and call timestamps, with rolling 24-hour cleanup during quota reservations. Session validity is 24 hours. Provider/hosting retention has not been independently confirmed. Keys stay server-side; same-origin, CSRF, quotas, concurrency and output validation mitigate misuse.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Questionnaire responses are unverified self-report, not validated aptitude, certified skill, hiring suitability or job-loss probability. Editorial matching and effort estimates may be incomplete or biased. O*NET is US occupational evidence; ESCO is a European taxonomy; Malaysia-specific occupation coverage is limited. Dated snapshots and official-file extracts are not live vacancy feeds. Check statistical period and geography before comparison. Unconfigured, slow, rate-limited or invalid RunPod output can fail. Deterministic reports remain available and no automatic paid retry is performed. Allowed-ID/schema checks reduce fabricated references but do not prove semantic entailment. Human review remains necessary. SQLite quotas do not cap provider bills across instances/deployments. Browser storage may be unavailable or exposed on shared devices. Engineering fixtures do not establish population-level fairness, improved employment, or emissions savings. Production optional-AI status and scheduled update execution require separate verification.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☐ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☐ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

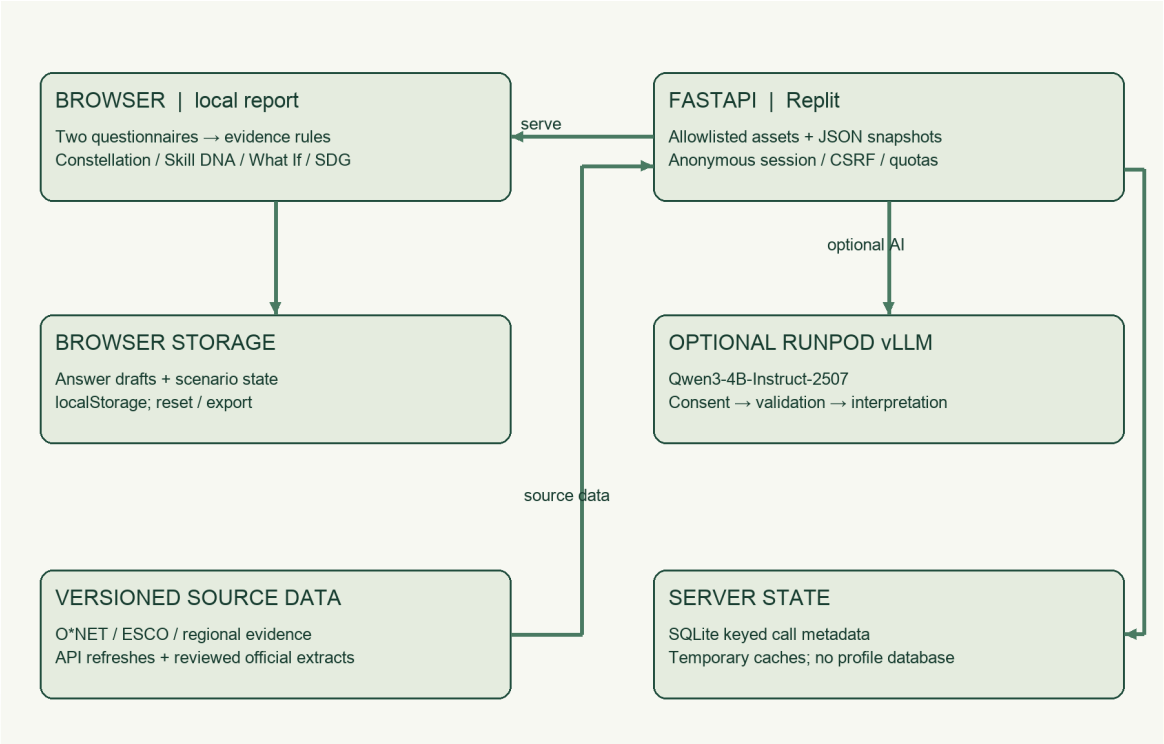
8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team’s knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Signature: /s/ XIE YIFENG (authorised typed electronic signature)

Signing date: 25 October 2026 (MYT)

Appendix A System architecture



Solid connections describe implemented source components. Optional inference is a separate request after consent; it does not replace the base report. External data is versioned and scoped; reviewed document extracts are not presented as live APIs.

Frontend: native ES modules. Server: Python/FastAPI. Replit configuration: cloudrun target, scripts/replit_start.sh, local port 8000 exposed as port 80. Public URL availability is verified; exact production commit and current model availability are not.

Appendix B Sources and evidence boundaries

[1] TalentCorp, Impact Study of AI, Digital and Green Economy on the Malaysian Workforce, Phase 1 scope: 10 sectors, 949 roles; approximately 620,000 employees (18% in analysed sectors) expected to be significantly impacted. https://www.talentcorp.com.my/impact-study/

[2] World Economic Forum, Future of Jobs Report 2025. Employer survey outlook: 39% of existing skill sets expected to change or become outdated, 2025-2030. Global survey; not a Malaysian job-loss estimate. https://www.weforum.org/publications/the-future-of-jobs-report-2025/

[3] United Nations, SDG targets 4.4, 8.2, 8.5 and 9.5. Proposed contribution pathways; no national indicator changes claimed. https://sdgs.un.org/goals

[4] O*NET Database 31.0, USDOL/ETA. Selected occupation/task/skill records. CC BY 4.0; adaptations and translations by Nexus are not endorsed by USDOL/ETA. https://www.onetcenter.org/license_db.html

[5] European Commission ESCO classification. Selected occupation/skill relationships and reviewed contextual links; not job-demand statistics. https://esco.ec.europa.eu/en/use-esco

[6] Product source inspected from a read-only GitHub main ZIP on 25 October 2026 MYT. Main source and historical documentation can differ; no live commit identity is asserted. https://github.com/xieyifeng24/AI-Career-Navigator

[7] Live prototype opened successfully in Chrome on 25 October 2026 MYT. Exact deployment SHA and production optional-AI configuration remain unverified. https://ai-career-navigator-xyf20070924.replit.app/

Source files: prototype/combined/current-analysis.js; starting-point.js; intelligence.js; pages/; src/career/api.py, access.py, regional.py, regional_ai.py and sources.py; requirements.txt; .replit. The downloaded code was inspected without running or modifying product code. Product acceptance results quoted in repository documents are team records, not tests rerun during this handoff.

Knowledge snapshot: knowledge-20261024T081728Z-d5b9e134322f0749; verified 24 October 2026, 16:17 MYT. Regional snapshot: regional-5b92e0e0305b4a1a; checked 24 October 2026, 10:13 MYT. Snapshot times are not the statistical reference periods.

Appendix C Complete resource inventory

Python distribution versions below are the project requirements.txt pins, not a claim that this handoff installed or verified the production environment. Licence families below require the team final package/notice check before attestation.

annotated-doc 0.0.5 | MIT

annotated-types 0.8.0 | MIT

anyio 4.15.1 | MIT

beautifulsoup4 4.15.0 | MIT

certifi 2026.10.19 | MPL-2.0

click 8.5.0 | BSD-3-Clause

fastapi 0.156.4 | MIT

h11 0.16.0 | MIT

httpcore 1.0.9 | BSD-3-Clause

httpx 0.28.1 | BSD-3-Clause

idna 3.20 | BSD-3-Clause

opentelemetry-api 1.45.1 | Apache-2.0

pydantic 2.14.1 | MIT

pydantic_core 2.50.1 | MIT

python-dotenv 1.2.4 | BSD-3-Clause

soupsieve 3.0.3 | MIT

starlette 1.9.0 | BSD-3-Clause

typing-inspection 0.4.4 | MIT

typing_extensions 4.16.0 | PSF-2.0

uvicorn 0.54.0 | BSD-3-Clause

Runtime/tooling: Python 3.12; native browser HTML/CSS/JavaScript and SVG/Canvas; SQLite standard library; Node.js for build/acceptance scripts; GitHub Actions checkout/setup-python and related workflow actions. No React, Vue, Firebase, Supabase, or separate cloud profile database was established from the inspected default implementation.

Current preparation tools: OpenAI Codex/ChatGPT interface, official-template OOXML editing, python-docx, Pillow and bundled LibreOffice for documents; OpenAI Artifact Tool for PPTX. These prepare submission material and are not production AI models.

Appendix D Source and genuine use evidence

O*NET: https://www.onetcenter.org/database.html ; https://www.onetcenter.org/license_db.html

ESCO: https://esco.ec.europa.eu/en/use-esco/use-esco-services-api

DOSM: https://storage.dosm.gov.my/labour/wages_occupation.csv ; MASCO: https://myjobprofile.dosm.gov.my/bda-myjobprofile/job/2512/software-developers

MyCOL: TalentCorp MyMahir Malaysia Critical Occupations List 2024/2025; exact official PDF URL is retained in data/regional/current.json.

BLS: https://www.bls.gov/emp/tables/occupational-projections-and-characteristics.htm ; reviewed table, not live API.

World Bank: https://api.worldbank.org/v2/country/MYS;USA/indicator/SL.UEM.TOTL.ZS?format=json

MYFutureJobs / direct ILOSTAT: not integrated as approved datasets. Source check times differ from reference periods. Retained English official extracts are labelled in localised reports.

Replit proof: live public deployment URL in Section 1, opened in Chrome on 25 October 2026 MYT; repository docs/evidence/replit-autoscale-published.png and .replit provide historical publication/configuration evidence. This does not certify exact current deployment SHA or sponsored billing.

RunPod proof: repository docs/evidence/runpod-smoke.json, runpod-semantic-review.json and runpod-gpu-log.png; runpod/README.md records the Qwen model and worker revision. These are inherited team evidence, not inference rerun during this handoff. Current production optional-AI availability, worker image digest and sponsor-credit classification remain unverified.

Evidence scope: Replit and RunPod use is documented. Workshop attendance, individual IP agreements, sponsor-funded balance and full licence audit are not certified by these records.

Replit proof of use

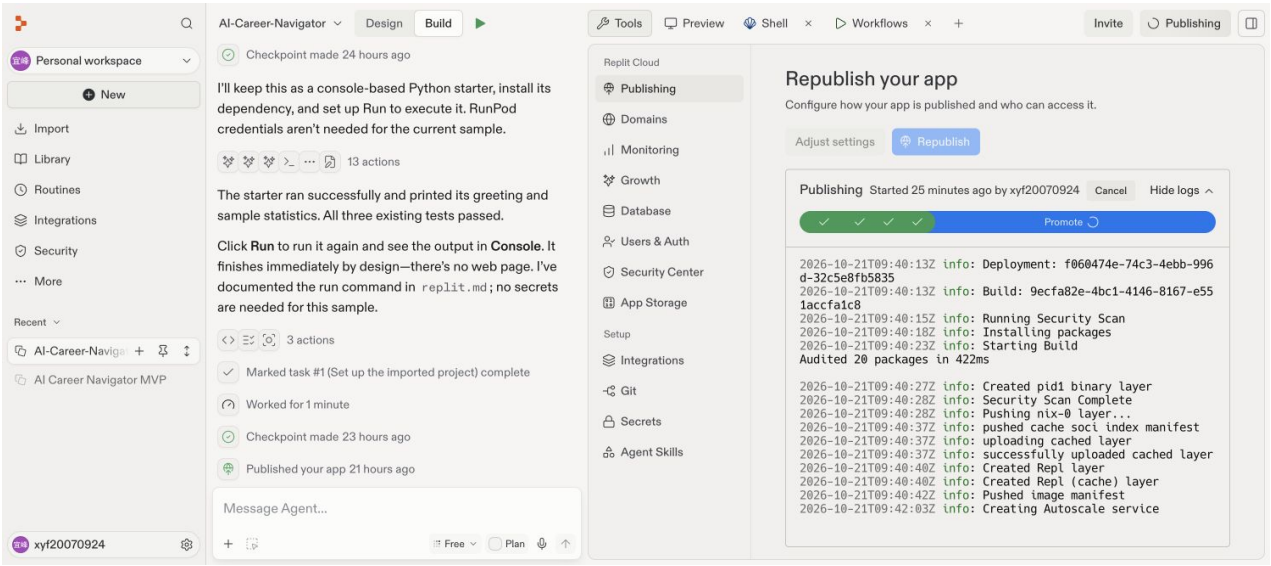
SDC Hackathon 2026 | Nexus | AI Career Navigator

Prepared 25 October 2026 MYT. This records genuine existing evidence; no new paid inference or deployment was performed for this document.

Public prototype: https://ai-career-navigator-xyf20070924.replit.app/ . Opened successfully in Chrome on 25 October 2026 MYT; homepage shows the AI Career Navigator brand, language control and assessment entry.

Project: https://replit.com/@xyf20070924/AI-Career-Navigator . Current repository configuration uses a Python 3.12 / FastAPI application, scripts/replit_start.sh, cloudrun deployment target, local port 8000 and external port 80.

Evidence boundary: The source screenshot below records deployment activity on 21 October 2026. Its progress state is not presented as a current successful-release receipt. Current URL availability was independently observed; exact running commit, deployment plan, region and sponsor-credit consumption were not independently verified.



Original repository evidence, docs/evidence/replit-autoscale-published.png. Caption intentionally distinguishes an in-progress publication log from observed current public availability.

Public source evidence: https://github.com/xieyifeng24/AI-Career-Navigator/tree/main/docs/evidence

RunPod proof of use

SDC Hackathon 2026 | Nexus | AI Career Navigator

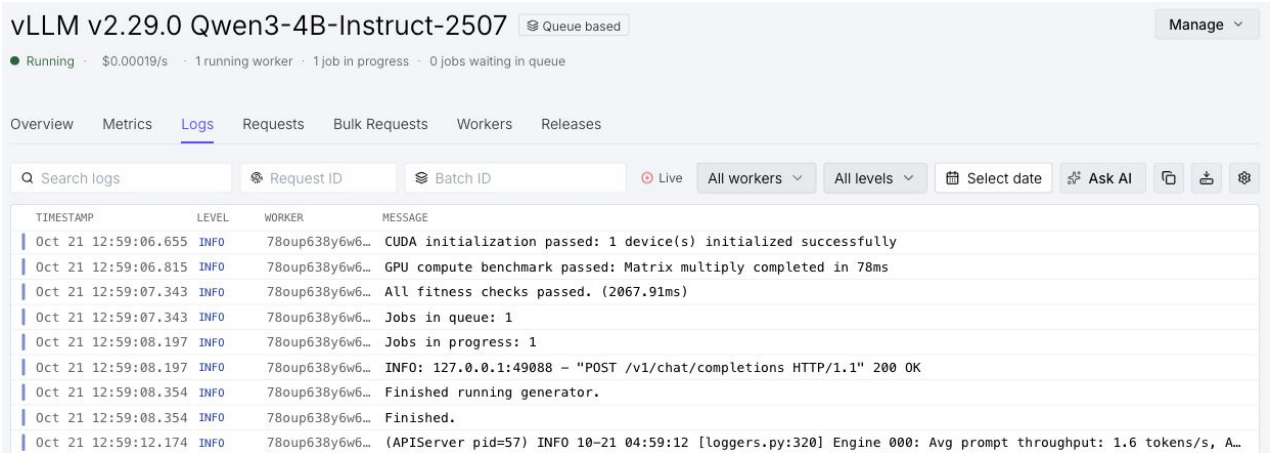
Prepared 25 October 2026 MYT. This records genuine existing evidence; no new paid inference or deployment was performed for this document.

Provider and workload: RunPod Serverless, official vLLM worker v2.29.0; Qwen/Qwen3-4B-Instruct-2507. Model revision: cdbee75f17c01a7cc42f958dc650907174af0554. Pretrained text generation; no fine-tuning is declared in the inspected code.

Recorded genuine GPU smoke inference, 21 October 2026: endpoint cckbk8134lpxf8; request chatcmpl-a71573c2671223a6; completion stopped normally; 21 input tokens + 10 output tokens; duration 117.6 seconds. The returned ready response proves a smoke inference, not career-report quality.

Full-report review boundary: request chatcmpl-bec48287722f94a5 completed in the historical record (75.67 seconds; 11,043 tokens), but semantic review blocked that release. This is disclosed rather than represented as an accepted final career report. See runpod-smoke.json and runpod-semantic-review.json.

Current implementation: server-only optional inference with consent, schema/evidence validation and core-report fallback. Existing use evidence is inherited from the team repository; this handoff did not rerun inference or verify current production endpoint configuration. Sponsor-credit status and worker image digest remain unverified.



Original repository GPU log, docs/evidence/runpod-gpu-log.png: CUDA initialisation and POST /v1/chat/completions HTTP 200. No API key is present in this evidence.

Public source evidence: https://github.com/xieyifeng24/AI-Career-Navigator/tree/main/docs/evidence